Anthopleura anthracene gene expression

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1 Data preparation

The updated workbook contains 19 gene columns, but KDM4A occurs twice and the two columns contain identical values. KDM4A was included in both the hypoxia and chromatin-remodeling plots in the workbook. The duplicate is removed here so that KDM4A does not receive twice the weight of every other gene during clustering.

Table 1: Data checks for the updated workbook

gene.columns.in.workbook	unique.genes.used	exact.duplicate.removed
19	18	KDM4A

The three rows for each treatment are the reported mean, minimum, and maximum, in that order. For the exploratory statistical analysis below, I assume that the minimum and maximum are two observations from exactly three independent replicates, and that the mean was calculated from those same three observations on the reported log2 scale. The unreported third observation is reconstructed from these values. This assumption should be confirmed before the results are used as final inferential statistics.

Converting data to long format

1.1 Reconstructing the triplicates

For three observations, the unreported value can be calculated as

$$x_3 = 3\bar{x} - x_{minimum} - x_{maximum}.$$

Table 2: Validation of reconstructed observations

gene.treatment.groups	reconstructed.observations	reconstructed.values.within.reported.range
108	108	108

2 Plots

2.1 Heatmap

First, I am creating a summary of gene expression by treatment for the heatmap.

Table 3: Mean log2 expression values used in the heatmap

	CNTL	20 ppb	24hr	96hr	7 day	Hypox
Ar	0	0.279	0.169	-0.261	-0.074	0.374
ARNT	0	0.361	0.206	-0.312	-0.281	0.228
HIF1A	0	-0.088	-1.012	-1.548	-0.873	0.222
HSP90	0	-0.737	0.368	-0.839	-0.704	-1.487
CYP1A1	0	-2.837	-0.642	-1.415	-1.308	-0.087
ABCB9	0	-0.178	1.320	1.500	1.181	-0.890
CAT	0	0.084	0.802	0.808	0.789	-0.693
AIMF1	0	0.634	0.706	0.162	-0.307	1.019
CASP8	0	-0.624	-0.286	-0.355	-0.331	-1.216
DAPK3	0	1.715	-0.341	1.046	1.325	0.395
XKR4	0	-1.454	0.593	0.141	0.123	-2.023
ARSB	0	-1.029	-1.415	-1.014	-1.065	0.725
KDM4A	0	-1.377	1.994	-0.783	-0.452	-1.282
SLC2A8	0	-2.108	-3.184	-4.506	-4.756	-0.856
SMYD	0	-0.852	0.386	-0.840	-0.749	0.061
SETDB1	0	-1.086	1.236	0.764	0.555	-1.101
SAP130	0	-1.326	0.566	-0.644	-0.444	-0.641
KAT5	0	-0.190	0.797	0.199	-0.254	-0.548

And next making the heatmap itself. The values are already log2 expression relative to mean control, so no additional row scaling is applied.

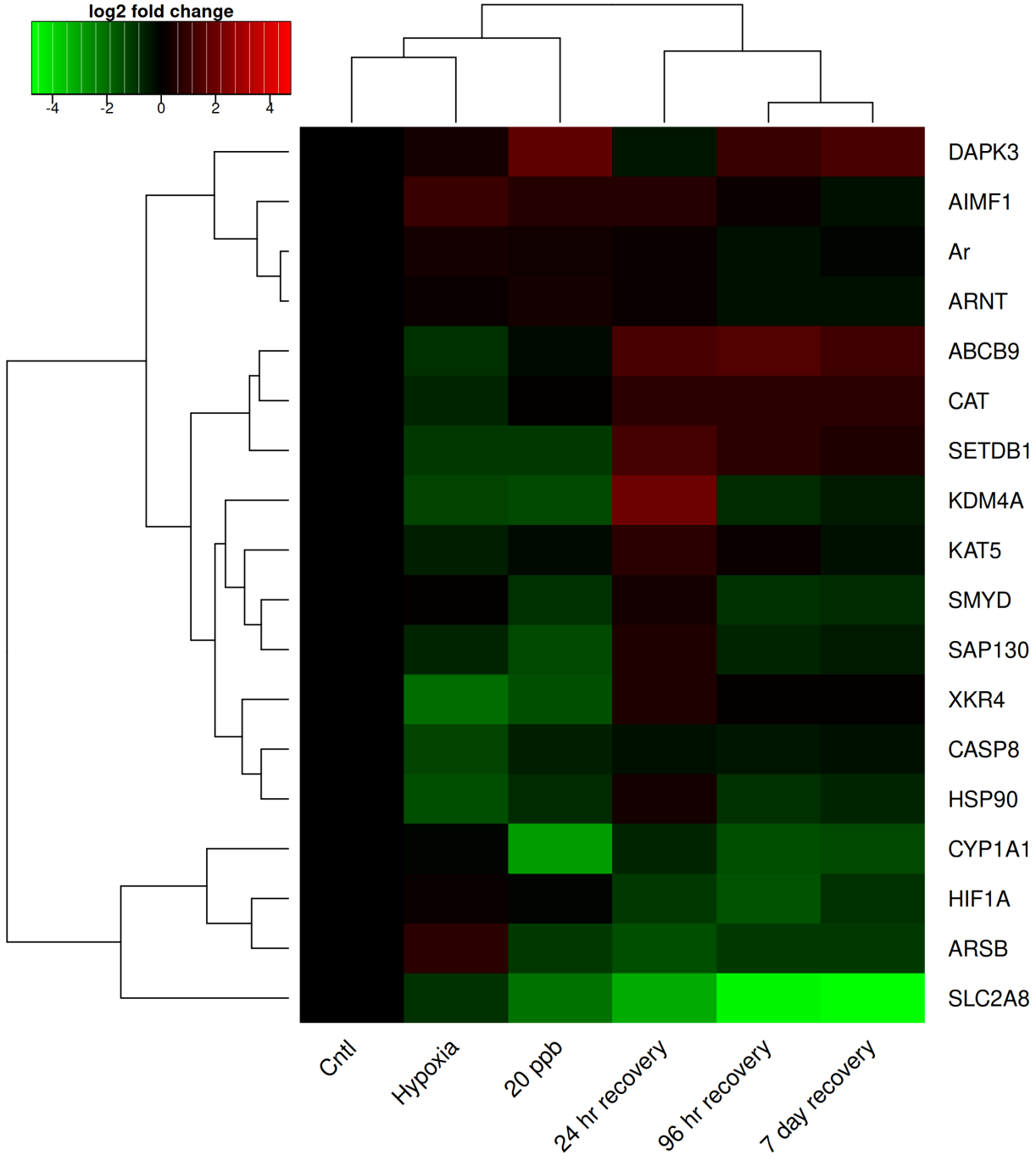


Figure 1: Clustered heatmap of mean expression for the six treatment groups.

3 Treatment comparisons

The comparisons are each recovery period versus both the control and 20 ppb treatments. The table below reports those differences in mean log2 expression. Positive values indicate greater expression in the recovery treatment than in the reference treatment.

Table 4: Descriptive recovery contrasts. These are effect sizes, not inferential tests or p-values.

Gene.ID	recovery	mean.expression	difference.from.control	difference.from.20ppb
ABCB9	24 hr recovery	1.320	1.320	1.498
ABCB9	96 hr recovery	1.500	1.500	1.678
ABCB9	7 day recovery	1.181	1.181	1.359
AIMF1	24 hr recovery	0.706	0.706	0.072
AIMF1	96 hr recovery	0.162	0.162	-0.472
AIMF1	7 day recovery	-0.307	-0.307	-0.941
ARNT	24 hr recovery	0.206	0.206	-0.155
ARNT	96 hr recovery	-0.312	-0.312	-0.673
ARNT	7 day recovery	-0.281	-0.281	-0.642
ARSB	24 hr recovery	-1.415	-1.415	-0.386
ARSB	96 hr recovery	-1.014	-1.014	0.015
ARSB	7 day recovery	-1.065	-1.065	-0.036
Ar	24 hr recovery	0.169	0.169	-0.110
Ar	96 hr recovery	-0.261	-0.261	-0.540
Ar	7 day recovery	-0.074	-0.074	-0.353
CASP8	24 hr recovery	-0.286	-0.286	0.338
CASP8	96 hr recovery	-0.355	-0.355	0.269
CASP8	7 day recovery	-0.331	-0.331	0.293
CAT	24 hr recovery	0.802	0.802	0.718
CAT	96 hr recovery	0.808	0.808	0.724
CAT	7 day recovery	0.789	0.789	0.705
CYP1A1	24 hr recovery	-0.642	-0.642	2.195
CYP1A1	96 hr recovery	-1.415	-1.415	1.422
CYP1A1	7 day recovery	-1.308	-1.308	1.529
DAPK3	24 hr recovery	-0.341	-0.341	-2.056
DAPK3	96 hr recovery	1.046	1.046	-0.669
DAPK3	7 day recovery	1.325	1.325	-0.390
HIF1A	24 hr recovery	-1.012	-1.012	-0.924
HIF1A	96 hr recovery	-1.548	-1.548	-1.460
HIF1A	7 day recovery	-0.873	-0.873	-0.785
HSP90	24 hr recovery	0.368	0.368	1.105
HSP90	96 hr recovery	-0.839	-0.839	-0.102
HSP90	7 day recovery	-0.704	-0.704	0.033
KAT5	24 hr recovery	0.797	0.797	0.987
KAT5	96 hr recovery	0.199	0.199	0.389
KAT5	7 day recovery	-0.254	-0.254	-0.064
KDM4A	24 hr recovery	1.994	1.994	3.371
KDM4A	96 hr recovery	-0.783	-0.783	0.594
KDM4A	7 day recovery	-0.452	-0.452	0.925
SAP130	24 hr recovery	0.566	0.566	1.892
SAP130	96 hr recovery	-0.644	-0.644	0.682
SAP130	7 day recovery	-0.444	-0.444	0.882
SETDB1	24 hr recovery	1.236	1.236	2.322
SETDB1	96 hr recovery	0.764	0.764	1.850
SETDB1	7 day recovery	0.555	0.555	1.641
SLC2A8	24 hr recovery	-3.184	-3.184	-1.076
SLC2A8	96 hr recovery	-4.506	-4.506	-2.398
SLC2A8	7 day recovery	-4.756	-4.756	-2.648
SMYD	24 hr recovery	0.386	0.386	1.238
SMYD	96 hr recovery	-0.840	-0.840	0.011
SMYD	7 day recovery	-0.749	-0.749	0.103
XKR4	24 hr recovery	0.593	0.593	2.047
XKR4	96 hr recovery	0.141	0.141	1.595
XKR4	7 day recovery	0.123	0.123	1.577

4 One-way ANOVA and post-hoc tests

Under the triplicate reconstruction assumption, a separate one-way ANOVA is fit for each GOI using all six treatments. The Benjamini–Hochberg column adjusts the 18 omnibus ANOVA p-values for testing multiple genes.

Table 5: One-way ANOVA for each GOI. BH-adjusted p-values account for the 18 omnibus gene tests.

Gene.ID	treatment.df	residual.df	F.value	p.value	p.adjusted.BH	significant.BH
SLC2A8	5	12	99.1460	< 0.001	< 0.001	TRUE
KDM4A	5	12	71.7629	< 0.001	< 0.001	TRUE
HIF1A	5	12	25.0673	< 0.001	< 0.001	TRUE
CYP1A1	5	12	22.6909	< 0.001	< 0.001	TRUE
CAT	5	12	22.2622	< 0.001	< 0.001	TRUE
SETDB1	5	12	16.3111	< 0.001	< 0.001	TRUE
ABCB9	5	12	16.0208	< 0.001	< 0.001	TRUE
HSP90	5	12	13.8790	< 0.001	< 0.001	TRUE
AIMF1	5	12	13.1461	< 0.001	< 0.001	TRUE
SMYD	5	12	12.3673	< 0.001	< 0.001	TRUE
CASP8	5	12	10.9691	< 0.001	< 0.001	TRUE
XKR4	5	12	9.7894	< 0.001	< 0.001	TRUE
ARSB	5	12	8.6948	0.00110	0.00152	TRUE
DAPK3	5	12	8.5599	0.00118	0.00152	TRUE
SAP130	5	12	8.1670	0.00146	0.00175	TRUE
KAT5	5	12	7.1830	0.00252	0.00283	TRUE
Ar	5	12	2.8090	0.06624	0.07013	FALSE
ARNT	5	12	0.3830	0.85103	0.85103	FALSE

Tukey's honestly significant difference test is then used for post-hoc comparisons. Tukey-adjusted p-values control the family-wise error rate across all 15 treatment pairs within each GOI. The full set of comparisons is written to `posthoc_results_tukey_all.csv`; the table below contains the six recovery recovery comparisons against control and 20 ppb.

Table 6: Tukey HSD recovery comparisons. A positive difference means that expression is greater in recovery than in the reference treatment.

GOI	recovery	reference	Difference	Lower 95% CI	Upper 95% CI	Tukey p	Significant
ABCB9	24 hr	20 ppb	1.4980	0.3380	2.6580	0.00961	TRUE
ABCB9	24 hr	Cntl	1.3200	0.1600	2.4800	0.02289	TRUE
ABCB9	7 day	20 ppb	1.3590	0.1990	2.5190	0.01892	TRUE
ABCB9	7 day	Cntl	1.1810	0.0210	2.3410	0.04515	TRUE
ABCB9	96 hr	20 ppb	1.6780	0.5180	2.8380	0.00407	TRUE
ABCB9	96 hr	Cntl	1.5000	0.3400	2.6600	0.00952	TRUE
AIMF1	24 hr	20 ppb	0.0720	-0.5809	0.7249	0.99884	FALSE
AIMF1	24 hr	Cntl	0.7060	0.0531	1.3589	0.03157	TRUE
AIMF1	7 day	20 ppb	-0.9410	-1.5939	-0.2881	0.00419	TRUE
AIMF1	7 day	Cntl	-0.3070	-0.9599	0.3459	0.62526	FALSE
AIMF1	96 hr	20 ppb	-0.4720	-1.1249	0.1809	0.22068	FALSE
AIMF1	96 hr	Cntl	0.1620	-0.4909	0.8149	0.95521	FALSE
ARNT	24 hr	20 ppb	-0.1550	-2.3105	2.0005	0.99985	FALSE
ARNT	24 hr	Cntl	0.2060	-1.9495	2.3615	0.99942	FALSE
ARNT	7 day	20 ppb	-0.6420	-2.7975	1.5135	0.90912	FALSE
ARNT	7 day	Cntl	-0.2810	-2.4365	1.8745	0.99742	FALSE
ARNT	96 hr	20 ppb	-0.6730	-2.8285	1.4825	0.89210	FALSE
ARNT	96 hr	Cntl	-0.3120	-2.4675	1.8435	0.99579	FALSE
ARSB	24 hr	20 ppb	-0.3860	-1.7038	0.9318	0.91457	FALSE
ARSB	24 hr	Cntl	-1.4150	-2.7328	-0.0972	0.03295	TRUE
ARSB	7 day	20 ppb	-0.0360	-1.3538	1.2818	1.00000	FALSE
ARSB	7 day	Cntl	-1.0650	-2.3828	0.2528	0.14291	FALSE
ARSB	96 hr	20 ppb	0.0150	-1.3028	1.3328	1.00000	FALSE
ARSB	96 hr	Cntl	-1.0140	-2.3318	0.3038	0.17468	FALSE
Ar	24 hr	20 ppb	-0.1100	-0.7808	0.5608	0.99253	FALSE
Ar	24 hr	Cntl	0.1690	-0.5018	0.8398	0.95238	FALSE
Ar	7 day	20 ppb	-0.3530	-1.0238	0.3178	0.51806	FALSE
Ar	7 day	Cntl	-0.0740	-0.7448	0.5968	0.99883	FALSE
Ar	96 hr	20 ppb	-0.5400	-1.2108	0.1308	0.14530	FALSE
Ar	96 hr	Cntl	-0.2610	-0.9318	0.4098	0.77634	FALSE
CASP8	24 hr	20 ppb	0.3381	-0.2594	0.9355	0.44598	FALSE
CASP8	24 hr	Cntl	-0.2860	-0.8834	0.3114	0.60886	FALSE
CASP8	7 day	20 ppb	0.2931	-0.3044	0.8905	0.58608	FALSE
CASP8	7 day	Cntl	-0.3310	-0.9284	0.2664	0.46698	FALSE
CASP8	96 hr	20 ppb	0.2691	-0.3284	0.8665	0.66354	FALSE

Table 6: Tukey HSD recovery comparisons. A positive difference means that expression is greater in recovery than in the reference treatment. *(continued)*

GOI	recovery	reference	Difference	Lower 95% CI	Upper 95% CI	Tukey p	Significant
CASP8	96 hr	Cntl	-0.3550	-0.9524	0.2424	0.39751	FALSE
CAT	24 hr	20 ppb	0.7180	0.1021	1.3339	0.01955	TRUE
CAT	24 hr	Cntl	0.8020	0.1861	1.4179	0.00905	TRUE
CAT	7 day	20 ppb	0.7050	0.0891	1.3209	0.02204	TRUE
CAT	7 day	Cntl	0.7890	0.1731	1.4049	0.01019	TRUE
CAT	96 hr	20 ppb	0.7240	0.1081	1.3399	0.01850	TRUE
CAT	96 hr	Cntl	0.8080	0.1921	1.4239	0.00857	TRUE
CYP1A1	24 hr	20 ppb	2.1950	1.1406	3.2494	< 0.001	TRUE
CYP1A1	24 hr	Cntl	-0.6420	-1.6964	0.4124	0.37364	FALSE
CYP1A1	7 day	20 ppb	1.5290	0.4746	2.5834	0.00399	TRUE
CYP1A1	7 day	Cntl	-1.3080	-2.3624	-0.2536	0.01280	TRUE
CYP1A1	96 hr	20 ppb	1.4220	0.3676	2.4764	0.00698	TRUE
CYP1A1	96 hr	Cntl	-1.4150	-2.4694	-0.3606	0.00724	TRUE
DAPK3	24 hr	20 ppb	-2.0560	-3.3562	-0.7558	0.00197	TRUE
DAPK3	24 hr	Cntl	-0.3410	-1.6412	0.9592	0.94412	FALSE
DAPK3	7 day	20 ppb	-0.3900	-1.6902	0.9102	0.90673	FALSE
DAPK3	7 day	Cntl	1.3250	0.0248	2.6252	0.04491	TRUE
DAPK3	96 hr	20 ppb	-0.6690	-1.9692	0.6312	0.54014	FALSE
DAPK3	96 hr	Cntl	1.0460	-0.2542	2.3462	0.14571	FALSE
HIF1A	24 hr	20 ppb	-0.9236	-1.5845	-0.2627	0.00532	TRUE
HIF1A	24 hr	Cntl	-1.0116	-1.6725	-0.3507	0.00258	TRUE
HIF1A	7 day	20 ppb	-0.7850	-1.4459	-0.1241	0.01725	TRUE
HIF1A	7 day	Cntl	-0.8730	-1.5339	-0.2121	0.00814	TRUE
HIF1A	96 hr	20 ppb	-1.4600	-2.1209	-0.7991	< 0.001	TRUE
HIF1A	96 hr	Cntl	-1.5480	-2.2089	-0.8871	< 0.001	TRUE
HSP90	24 hr	20 ppb	1.1050	0.2663	1.9437	0.00830	TRUE
HSP90	24 hr	Cntl	0.3680	-0.4707	1.2067	0.68556	FALSE
HSP90	7 day	20 ppb	0.0330	-0.8057	0.8717	0.99999	FALSE
HSP90	7 day	Cntl	-0.7040	-1.5427	0.1347	0.12109	FALSE
HSP90	96 hr	20 ppb	-0.1020	-0.9407	0.7367	0.99814	FALSE
HSP90	96 hr	Cntl	-0.8390	-1.6777	-0.0003	0.04989	TRUE
KAT5	24 hr	20 ppb	0.9867	0.1644	1.8089	0.01611	TRUE
KAT5	24 hr	Cntl	0.7970	-0.0252	1.6193	0.05939	FALSE
KAT5	7 day	20 ppb	-0.0644	-0.8867	0.7579	0.99978	FALSE
KAT5	7 day	Cntl	-0.2540	-1.0763	0.5682	0.89614	FALSE
KAT5	96 hr	20 ppb	0.3887	-0.4335	1.2110	0.62035	FALSE
KAT5	96 hr	Cntl	0.1991	-0.6231	1.0214	0.95943	FALSE
KDM4A	24 hr	20 ppb	3.3710	2.6737	4.0683	< 0.001	TRUE
KDM4A	24 hr	Cntl	1.9940	1.2967	2.6913	< 0.001	TRUE
KDM4A	7 day	20 ppb	0.9250	0.2277	1.6223	0.00790	TRUE
KDM4A	7 day	Cntl	-0.4520	-1.1493	0.2453	0.31433	FALSE
KDM4A	96 hr	20 ppb	0.5940	-0.1033	1.2913	0.11330	FALSE
KDM4A	96 hr	Cntl	-0.7830	-1.4803	-0.0857	0.02494	TRUE
SAP130	24 hr	20 ppb	1.8920	0.8228	2.9612	< 0.001	TRUE
SAP130	24 hr	Cntl	0.5660	-0.5032	1.6352	0.51225	FALSE
SAP130	7 day	20 ppb	0.8820	-0.1872	1.9512	0.13084	FALSE
SAP130	7 day	Cntl	-0.4440	-1.5132	0.6252	0.72958	FALSE
SAP130	96 hr	20 ppb	0.6820	-0.3872	1.7512	0.32930	FALSE
SAP130	96 hr	Cntl	-0.6440	-1.7132	0.4252	0.38425	FALSE
SETDB1	24 hr	20 ppb	2.3220	1.1710	3.4730	< 0.001	TRUE
SETDB1	24 hr	Cntl	1.2360	0.0850	2.3870	0.03294	TRUE
SETDB1	7 day	20 ppb	1.6410	0.4900	2.7920	0.00456	TRUE
SETDB1	7 day	Cntl	0.5550	-0.5960	1.7060	0.60214	FALSE
SETDB1	96 hr	20 ppb	1.8500	0.6990	3.0010	0.00171	TRUE
SETDB1	96 hr	Cntl	0.7640	-0.3870	1.9150	0.29273	FALSE
SLC2A8	24 hr	20 ppb	-1.0761	-1.9977	-0.1546	0.01933	TRUE
SLC2A8	24 hr	Cntl	-3.1840	-4.1056	-2.2624	< 0.001	TRUE
SLC2A8	7 day	20 ppb	-2.6481	-3.5697	-1.7266	< 0.001	TRUE
SLC2A8	7 day	Cntl	-4.7560	-5.6776	-3.8344	< 0.001	TRUE
SLC2A8	96 hr	20 ppb	-2.3981	-3.3197	-1.4766	< 0.001	TRUE
SLC2A8	96 hr	Cntl	-4.5060	-5.4276	-3.5844	< 0.001	TRUE
SMYD	24 hr	20 ppb	1.2375	0.5021	1.9730	0.00116	TRUE
SMYD	24 hr	Cntl	0.3859	-0.3495	1.1213	0.52089	FALSE
SMYD	7 day	20 ppb	0.1029	-0.6326	0.8383	0.99641	FALSE
SMYD	7 day	Cntl	-0.7487	-1.4842	-0.0133	0.04516	TRUE

Table 6: Tukey HSD recovery comparisons. A positive difference means that expression is greater in recovery than in the reference treatment. *(continued)*

GOI	recovery	reference	Difference	Lower 95% CI	Upper 95% CI	Tukey p	Significant
SMYD	96 hr	20 ppb	0.0113	-0.7241	0.7467	1.00000	FALSE
SMYD	96 hr	Cntl	-0.8403	-1.5757	-0.1049	0.02230	TRUE
XKR4	24 hr	20 ppb	2.0470	0.4617	3.6323	0.00962	TRUE
XKR4	24 hr	Cntl	0.5930	-0.9923	2.1783	0.80181	FALSE
XKR4	7 day	20 ppb	1.5770	-0.0083	3.1623	0.05149	FALSE
XKR4	7 day	Cntl	0.1230	-1.4623	1.7083	0.99979	FALSE
XKR4	96 hr	20 ppb	1.5954	0.0101	3.1807	0.04823	TRUE
XKR4	96 hr	Cntl	0.1414	-1.4439	1.7267	0.99958	FALSE

These inferential results are provisional because they depend on the supplied minimum and maximum being actual observations from a triplicate and on the mean having been calculated from those same observations on the same scale. With only three observations per treatment, assumption checks also have limited power. The models should therefore be rerun using the original replicate-level measurements when those data become available.